# MALEESHA PIYUMINI PATHIRANA

**Software Engineering Intern**

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## SUMMARY

IT undergraduate at ITUM with hands-on experience in Computer Vision, Deep Learning, and Full-Stack development. Skilled in building end-to-end edge AI pipelines, computer vision models, and scalable backend architectures; seeking a Software Engineering Internship.

## TECHNICAL SKILLS

- **Languages:** Python, Java, C/Embedded C, JavaScript, SQL, Dart
- **AI / ML & Computer Vision:** OpenCV, MediaPipe, Scikit-learn, NLP, Deep Learning, Edge AI
- **Frameworks & Web:** Spring Boot, Node.js, React, REST APIs, Flutter
- **Databases & DevOps:** MySQL, PostgreSQL, Docker, Git, GitHub
- **Embedded & Tools:** ATmega32, Proteus, Linux, Postman

## EDUCATION

**National Diploma in Technology (NDT) in Information Technology** — **Expected: 2027**
Institute of Technology, University of Moratuwa (ITUM) — *CGPA: 3.31 / 4.00*

## PROJECTS

**Zero Waste – Smart Surplus Food Redistribution Platform (FYP)** *React, Spring Boot, MySQL, Docker*

- Designed modular enterprise backend microservices to redistribute surplus food and minimize commercial waste.
- Engineered real-time donor-receiver matching logic, automated reporting services, and containerized deployments via Docker.

**VigilDrive AI – Edge Vision & Driver Safety Ecosystem** *Python, OpenCV, MediaPipe, ATmega32, Flutter*

- Implemented computer vision pipelines using MediaPipe Face Mesh to monitor eye-closure (EAR) and head pose (solvePnP) for driver fatigue.
- Built a two-wheeler safety circuit in Proteus using ATmega32 to detect head-tilt fatigue and crash impacts via ADC sensors with hardware SOS interrupts.
- Streamed embedded sensor telemetry and real-time vision alerts directly to a cross-platform Flutter monitoring dashboard.

**AI-Powered Anonymous Crisis Intervention & Evidence Platform** *JavaScript, React, Node.js, Web Security*

- Created an encrypted platform for anonymous reporting and tamper-proof evidence preservation for cyberbullying victims.
- Ensured secure data integrity, cryptographic hashing, and user confidentiality across client-server interactions.

**AI-Powered Career Guidance & Skill-Gap Platform** *Python NLP, React, Node.js, REST APIs*

- Developed an NLP platform evaluating student profiles across 7+ industries to detect technical skill gaps.
- Generated automated, dynamic learning roadmaps based on semantic analysis and industry benchmarks.

## CERTIFICATIONS & LANGUAGES

- **Certifications:** AI/ML Engineer - Stage 1 (SLIIT) | Applied Data Science with Python (Simplilearn) | Python for Beginners (CODL, UoM) | Front-End Web Development (CODL, UoM) | German 1 (The Open University)
- **Languages:** English (Professional), Sinhala (Native)
- **Extracurricular:** Technical Presentation & Communication, Vocal Artistry & Media Reviews

## REFERENCES

**Mrs. D. A. S. Ruwanmini**
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